



MECH MAINT- BOILER Department
SIMHADRI



Ref: SMTTP/MECH MAINT-/2022-23/LMI/611938

Date:

20/11/2022(22:58:03)

Subject: Review of LMI "Maintenance of the Integrity of Boiler Support Systems"

OPERATION & MAINTENANCE

MECHANICAL MAINTENANCE

(PRESSURE PARTS)

REF.: SMPP/O&M/MM/PP/

Date: 20.11.2022

Subject: Review of LMI "Maintenance of the Integrity of Boiler Support Systems"

LMI No: LMI/OGN/OPS/MECH/016

The above approved LMI was prepared based on reference document - " **COS-ISO-00-OGN/OPS/MECH/016**" circulated by CC-OS. The envisaged changes in LMI are incorporated as per details at revision index of LMI.

Put up for approval please

Submitted by:

राहुल जैन - वरिष्ठ प्रबंधक

Rahul Jain - 013328 (SENIOR MANAGER ,MECH MAINT- BOILER) ,Simhadri
(RAHULJAIN@NTPC.CO.IN, 9415246668)

On: 20/11/2022(22:58:03) **Ref:** SMTTP/MECH MAINT-/2022-23/LMI/611938

The subject LMI with doc ref- LMI/OGN/OPS/MECH/016/ ISSUE-1; REV.-5.00; Nov'2022 has been revised after incorporating changes as per OGN doc with ref- COS-ISO-00-OGN/OPS/MECH/016.

May please accord approval for this revised version of LMI.

Submitted by:

सुरेश बाबु कोल्ला - उप महाप्रबंधक

Suresh Babu Kolla - 007980 (DY. GENERAL MANAGER ,MECH MAINT- BOILER)
(SBKOLLA@NTPC.CO.IN, 9490307556) ,Simhadri

On:26/11/2022(19:00:44) **Ref:**SMTTP/MECH MAINT-/2022-23/LMI/611938

1.Revision done as per clause 5.4.2 and 5.8.5 of nre OGN

2. Annexure -1 check list included in revised LMI

Revised LMI (attached doc NEW LMI) may please be approved.

Submitted by:

ओम प्रकाश - अपर महाप्रबंधक

Om Prakash - 007105 (ADDL. GENERAL MANAGER ,MECH MAINT- BOILER)
(OMPRAKASH08@NTPC.CO.IN, 9650994070) ,Simhadri

On:28/11/2022(09:11:47) **Ref:**SMTTP/MECH MAINT-/2022-23/LMI/611938

Submitted for approval pls

Submitted by:

झिमलि धर - अपर महाप्रबंधक

Jhimli Dhar - 006460 (ADDL. GENERAL MANAGER ,EEMG)
(JHIMLIDHAR@NTPC.CO.IN, 9650998085) ,Simhadri

On:29/11/2022(09:35:08) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/611938

New LMI (revised) may please be accorded approval.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

Mathew Eipe Kovoov - 005637 (GENERAL MANAGER ,MAINTENANCE)
(MEKOVOOR@NTPC.CO.IN, 9650994652) ,Simhadri

On:29/11/2022(11:12:09) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/611938

कृपया अनुमोदन प्रदान किया जाए।

May be approved please.

Submitted by:

राजेश कुमार झा - महाप्रबंधक

Rajesh Kumar Jha - 003323 (GENERAL MANAGER ,O & M)
(RKJHA02@NTPC.CO.IN, 9415342046) ,Simhadri

On:29/11/2022(11:52:08) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/611938

Approved & Submitted by:

गिरीश चन्द्र चौकसे - मुख्य महाप्रबंधक

Girish Chandra Choukse - 002907 (CHIEF GENERAL MANAGER ,HEAD OF PROJECT)
(GCCHOUKSE@NTPC.CO.IN, 9425178006) ,Simhadri

On:30/11/2022(12:24:58) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/611938

NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/MECH/016/

ISSUE NO.1; REVISION NO. 5.00; DATE: Nov'2022

**MAINTENANCE OF THE INTEGRITY
OF
BOILER SUPPORT SYSTEMS**

APPROVED FOR

***Implementation*-----**

HOP, SIMHADRI

Date-----

LMI APPROVAL DOCUMENTS

<u>1.0 TITLE OF LMI</u>	MAINTENANCE OF INTEGRITY OF BOILER SUPPORT SYSTEMS
<u>2.a) DOCUMENT CODE</u>	<u>LMI/OGN/OPS/MECH/016/</u>
b) REVISION/ISSUE NO.&DATE	<u>ISSUE-1; REV.-5.00; Nov'2022</u>
<u>3.ORIGINAL/REF.DOCUMENT</u>	COS-ISO-00-OGN/OPS/MECH/016 Rev. No. : 1 June 2020
<u>4.PREPARED BY:</u> Sr. MANAGER (MM-PP)	SIGNATURE-----
<u>5. CHECKED BY:</u> DGM(BMD/PP) AGM (BMD) GM (MTC.) GM (O&M)	SIGNATURE_____ SIGNATURE_____ SIGNATURE_____ SIGNATURE_____
<u>6. REVIEWED AND APPROVED BY</u>	SIGNATURE:----- HOP, SIMHADRI
<u>7.DATE OF ISSUE</u>	
<u>8.INDEX LIST</u>	ENCLOSED.
<u>9.DISTRIBUTION LIST</u>	ENCLOSED.

NTPC	LMI	SIMHADRI
TITLE: MAINTENANCE OF INTEGRITY OF BOILER SUPPORT SYSTEMS REV: 5.00 DATE: Nov'2022		
REF NO. LMI/OGN/OPS/MECH/016/		PAGE : 1 OF 8
<u>1.0 INDEX</u>		
<u>SECTION</u>	<u>TITLE</u>	<u>PAGE</u>
1.0	INDEX	1
2.0	INTRODUCTION	2
3.0	SCOPE	2
4.0	REFERENCE DOCUMENT	2
5.0	SUPERSEDED DOCUMENT	2
6.0	SPECIFIC INSTRUCTIONS	3
7.0	METHODOLOGY OF IMPLEMENTATION & RESPONSIBILITY CENTERS	3
8.0	EXCEPTION REPORTING	4
9.0	RESPONSIBILITY	4
10.0	RECORD KEEPING	5
11.0	REVIEW	5
12.0	REVISION INDEX	6
13.0	DISTRIBUTION LIST	7
14.0	COMPLIANCE RECORD	8

NTPC	LMI	SIMHADRI
TITLE: MAINTENANCE OF INTEGRITY OF BOILER SUPPORT SYSTEMS REV: 5.00 DATE: Nov'2022		
REF NO. LMI/OGN/OPS/MECH/016/		PAGE : 2 OF 8
<u>2.0 INTRODUCTION</u> The two-pass drum type boilers at Simhadri are hung from the top main suspension steel work supported by the main boiler house columns. Drums are supported by tie rods hung from the top main suspension steel work. The main boiler walls are hung from the top steel work. The radiant & convection tube bank assemblies together with the roof tubes are also hung via intermediate beam assemblies from the top main steel work. Economizers are supported from top main steel work & also by beams which span in the gas passes between enclosure front & rear walls. At the top of the boiler, the boiler inner casing forms a canopy roof enclosing the boiler drum, all riser pipes, headers & the inter stage pipe works & tubing above the furnace and enclosure roof tubes called dead space. The furnace and enclosure roof tubes are covered with a refractory layer & plate work, and this forms the primary seal between gas and dead space. All the boiler supports necessarily pass-through boiler top dead space, where they are subjected to loading and temperature. Support structures are also located in furnace nose and bottom dead spaces. The objective of this LMI is to maintain the integrity of boiler support structures located in dead spaces. <u>3.0 SCOPE</u> The scope of this LMI is to detail the methods that are to be adopted for the maintenance of structural integrity of boiler support structures. It refers essentially to the top main steel suspension steel work and the associated boiler support components and attachments located in the top dead space. <u>4.0 REFERENCE DOCUMENT</u> Operation guidance note OGN/OPS/MECH/016 circulated by operation services-NTPC corporate center. <u>5.0 SUPERSEDED DOCUMENT</u> MAINTENANCE OF INTEGRITY OF BOILER SUPPORT SYSTEMS REV: 2.00 DATE: APR'2013		

NTPC	LMI	SIMHADRI
TITLE: MAINTENANCE OF INTEGRITY OF BOILER SUPPORT SYSTEMS REV: 5.00 DATE:Nov'2022		
REF NO. LMI/OGN/OPS/MECH/016/		PAGE : 3 OF 8
<u>6.0 SPECIFIC INSTRUCTIONS</u>		
To achieve the objective of this LMI the following approach is to be followed :		
a) Regular monitoring of dead space dust levels. b) Regular removal of dead space dust accumulations. c) Control of dead space dust ingress d) Periodic inspection of dead space support components and necessary preventive actions.		
<u>7.0 METHODOLOGY OF IMPLEMENTATION AND RESPONSIBILITY CENTERS</u>		
a) Inspection of boiler dead space (top dead space and furnace nose dead space) to monitor the dust levels during shutdowns of the units for a time greater than 48 hrs. Allowable dust depth is 300 mm.		
Responsibility – HOS (PP) Frequency – during unit breakdowns.		
b) Removal of boiler dead space dust accumulations with proper safety precautions and application of flexible soft sealing compounds in more defect prone locations.		
Responsibility – HOS (PP) Frequency – during shutdowns as per need basis.		
c) Dust level survey to be carried out at the beginning of overhaul to determine local seal deterioration and for possible presence of furnace roof burn holes. Amount of dust collected from dead space during overhaul cleanup, to be weighed and to be logged along with boiler running hours since last clean up action.		
Responsibility – HOS.(PP) Frequency - During Overhaul		
d) Inspection of roof seal plate work and refractory and repair work as necessary.		
Responsibility – HOS.(PP) Frequency - During Overhaul		
e) Inspection of boiler support components as detailed below after dust clearance of dead space (bottom and nose) for structural integrity/looseness. If the lower end of a rod is hidden in lagging, lagging may be removed to check the connection. I. Inside dead space a) Drum straps b) Supports of pipe work associated with drum c) Main boiler wall supports d) Radiant and convective tube bank supports		

NTPC	LMI	SIMHADRI								
TITLE: MAINTENANCE OF INTEGRITY OF BOILER SUPPORT SYSTEMS REV: 5.00 DATE: Nov’2022										
REF NO. LMI/OGN/OPS/MECH/016/		PAGE : 4 OF 8								
e) Headers, mainstream and hot reheat line supports f) Furnace and enclosure roof seals (radiant and convective tube bank penetrations) g) Roof tube supports including random sample of roof tube connection lugs II. Components at canopy roof level a) Canopy roof plate work/steel work b) Support connections to canopy roof plate work /steel work c) Seal pots where sling rods pass through canopy roof III. Components above canopy roof a) Parts of sling line above canopy roof, including connections to top main steel work frame b) Top frame steel work c) Condition of washer/packer plate under the main tie rod nut. Responsibility – HOS(PP) Frequency - During Overhaul f) Inspection of support suspension rods for necking and reduction of diameter on account of dew point corrosion during overhaul. Responsibility – HOS(PP) Frequency - During Overhaul Repair and rectification of defects observed including tightening of loose rods to be done after consultation with corporate engineering & OEM. Where it is necessary to measure the existing loads in a sling line, a check weighing method, using a series of hydraulic jacks dynamometers may be used. If the Tie rod support frame bottom side fitted with lock nut the same should have gap from the structure to ensure contraction during cooling.										
8.0 EXCEPTION REPORTING										
The exceptions encountered during inspection are to be reported as per the following format and to be taken up with corporate engineering & manufacturer										
----- HOD(BMD)										
<table><tr><td>Description of exception</td><td>Unit No.</td><td>Location of exception</td><td>Remarks</td></tr><tr><td></td><td></td><td></td><td></td></tr></table>			Description of exception	Unit No.	Location of exception	Remarks				
Description of exception	Unit No.	Location of exception	Remarks							
9.0 RESPONSIBILITIES										
The responsibility of carrying out different tasks in relation to inspection and maintenance of integrity of boiler support systems as detailed in LMI are with HOD (BMD)										

NTPC	LMI	SIMHADRI
TITLE: MAINTENANCE OF INTEGRITY OF BOILER SUPPORT SYSTEMS REV: 5.00 DATE: Nov'2022		
REF NO. LMI/OGN/OPS/MECH/016/		PAGE : 5 OF 8
<u>10.0. RECORD KEEPING</u> The records to be maintained under this LMI include a) compliance record of the activities detailed under this LMI b) reports of survey and rectification carried out c) Exception reporting The above records are to be maintained by HOD(BMD). <u>11.0.REVIEW</u> This document shall be reviewed by HOP (Simhadri) on TWO yearly basis.		

NTPC		LMI		SIMHADRI
TITLE: MAINTENANCE OF INTEGRITY OF BOILER SUPPORT SYSTEMS REV: 5.00 DATE: Nov'2022				
REF NO. LMI/OGN/OPS/MECH/016/			PAGE : 6 OF 8	
12. REVISION INDEX (For changes/revision)				
Sl. No.	Rev. no.	Date	Details	Approved by
01	001	29.01.2010	i) Approval routing changed at "LMI Approval Document"	
			ii) Reviewing authority changed from DGM (MM) to GM I/c	
			iii) Distribution List	
02	002	30.04.2013	i) Responsibility changed from DGM (PP) to HOS (Pressure Parts) at Responsibilities	
			ii) Inclusion of U#3 & 4 in compliance record	
			iii) Distribution List	
03	003	16.01.2020	i) Approval routing changed at "LMI Approval document"	
			ii) Review Frequency changed from "three years" to "TWO years"	
			iii) Distribution List	
04	004	24.06.2020	i) Last Paragraph of the 7 f) has been added as per clause 5.4.3 (ii) of COS-ISO-00-GN/OPS/MECH/016 Rev. No.: 1 Date: June 2020	
			ii) Distribution List	
05	004	13.07.2022	No Changes Done	
06	005	20.11.2022	Changes done in bold as per COS-ISO-00-OGN/OPS/MECH/016 Rev 2, Oct 2022	

[illegible]

NTPC	14.0 COMPLIANCE RECORD	SIMHADRI
TITLE: MAINTENANCE OF INTEGRITY OF BOILER SUPPORT SYSTEMS REV:5.00 DATE: Nov'2022		
REF NO. LMI/OGN/OPS/MECH/016/		PAGE : 8 OF 8
UNIT NO.	ACTIVITY	COMPLIANCE
I/II/III/IV		

Annexure-1 Checklist for Integrity of Boiler support Systems

<u>SI No</u>	<u>Component /Area</u>	<u>Observations</u>	<u>Action taken</u>
1	Drum straps		
2	Support of pipe associated with drum		
3	Main boiler wall supports		
4	Radiant and convective tube bank supports		
5	Headers (CRH/HRH/MS) line supports		
6	Radiant and convective tube bank penetrations seal bands		
7	Furnace and enclosure roof refractory covering and plate work.		
8	Roof tube supports		
9	Roof tubes (Sagged roof tubes)		
10	Gooseneck bottom dead chamber support structure		
11	S-Panel Bottom dead chambers support structure		
12	Expansion bellow at front (At roof tube entry)		
13	Condition of expansion bellow at both sides		
14	Canopy roof platework/steelwork		
15	Seal pots where sling rods pass through the canopy roof		
16	Parts of sling line above canopy roof, including connection to the top steel work frame		
17	Top frame steel work		
18	Condition of washer/packers plate under the main tie rod nut		

Signature of BMD

Signature of Operation

